

Missing values

Facundo Muñoz

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Contents

The handling of missing values (i.e. NA) depends on *where* they are.

Missing response

It is perfectly valid to have missing values in the dependent variable. There is no need of removing those individuals from the dataset. Furthermore, including them will yield predictions for their phenotype, based on the predictive variables.

```
library(breedR)

N <- 1e3
x <- rep(1:4, each = N/4)
dat <- data.frame(y = x + rnorm(N),
                  x = factor(letters[x]))
dat$y[1] <- NA
head(dat)

##           y x
## 1         NA a
## 2 0.69274277 a
## 3 0.09790199 a
## 4 1.62706874 a
## 5 2.12035503 a
## 6 3.12721355 a

res <- remlf90(y ~ x, data = dat)

## The predicted phenotype for y[1] is the estimated effect
## of the corresponding level of x
fitted(res)[1] == fixef(res)$x['a']

##      1
## TRUE
```

Missing value for a fixed effect

This is not allowed, as it would yield an underdetermined system of equations. **breedR** issues an error if missing values are detected.

```

N <- 1e3
x <- rep(1:4, each = N/4)
dat <- data.frame(y = x + rnorm(N),
                  x = factor(letters[x]))
dat$x[c(1, 3, 5)] <- NA
head(dat)

```

```

##           y      x
## 1 0.3211924 <NA>
## 2 1.5743127    a
## 3 0.2954855 <NA>
## 4 0.4660159    a
## 5 1.7743846 <NA>
## 6 0.5243786    a

```

```

res <- remlf90(y ~ x, data = dat)

```

```

## Error in progsf90(mf, weights = weights, effects, opt = union("sol se", :
## Missing values in covariates are not allowed
## check individuals: 1, 3, 5

```

Idem for a regression variable.

```

N <- 1e3
x <- runif(N)
dat <- data.frame(y = 1 + 2*x + rnorm(N),
                  x = x)
dat$x[c(1, 3, 5)] <- NA
head(dat)

```

```

##           y      x
## 1  3.0572315    NA
## 2  2.8322030 0.5910449
## 3  1.8304948    NA
## 4  1.4058794 0.8450186
## 5 -0.9665408    NA
## 6  4.8346455 0.8146597

```

```

res <- remlf90(y ~ x, data = dat)

```

```

## Error in progsf90(mf, weights = weights, effects, opt = union("sol se", :
## Missing values in covariates are not allowed
## check individuals: 1, 3, 5

```

Missing value for a random effect

These **are** allowed. The incidence matrix will have a row of zeros for the corresponding individual.

```

N <- 1e3
N.blk <- 20
blk.effects <- rnorm(N.blk, sd = 2)
blk.idx <- sample(seq_len(N.blk), N, replace = TRUE)
dat <- data.frame(y = 1 + blk.effects[blk.idx] + rnorm(N),
                  blk = factor(blk.idx))
dat$blk[1] <- NA
head(dat)

```

```
##           y blk
## 1  2.0896693 <NA>
## 2  1.6667562  13
## 3  1.3489663  14
## 4  5.0050920  19
## 5  0.2552874  14
## 6 -0.8297340  20

res <- remlf90(y ~ 1, random = ~ blk, data = dat)

sum(model.matrix(res)$blk[1,])

## [1] 0
```

As a consequence, the predicted phenotype will be based on the remaining available effects. In this case, the global mean.

```
fitted(res)[1] == fixef(res)$Intercept[1]

##      1
## TRUE
```

The spatial block effect is another way of writing the previous experiment. So it works in the same way.

```
coord <- expand.grid(row = 1:20, col = 1:50)
res <- remlf90(y ~ 1,
              spatial = list(model = 'blocks',
                             coord = coord,
                             id     = 'blk'),
              data = dat)

c(sum(model.matrix(res)$spatial[1,]) == 0,
  fitted(res)[1] == fixef(res)$Intercept[1])

##      1
## TRUE TRUE
```

However, the empirical residuals of the individuals with missing values of the random effects will have an increased variance. We can show that by replicating the previous experiment and computing the variance of the residual for the first observation.

```
resid_sample <- replicate(1e2, sample_first_residual())
var(resid_sample)

## [1] 3.171813
```

This can be important when fitting several random effects. See below.

Missing values in genetic effects

For an additive genetic effect, the relationship between individuals is given in the pedigree. It is legitimate not knowing the relatives for some individual. This is what happens with founders, for example.

Use NA for unknown relatives. If both are unknown (e.g. founders), the genetic effect (Breeding Value) will be predicted based on its phenotype, the other effects, and the estimated heritability.

```
dat <- breedR.sample.phenotype(
  fixed = c(mu = 10, x = 2),
  genetic = list(model = 'add_animal',
```

```

      Nparents = c(10, 10),
      sigma2_a = 2,
      check.factorial = FALSE),

  N = 1e3)
head(dat)

##    self sire dam X.mu      X.x      BV      resid phenotype
## 1     1   NA  NA   1 0.8214305 -1.8479332  0.73623198  10.53116
## 2     2   NA  NA   1 0.9612078  1.3318850 -0.07417135  13.18013
## 3     3   NA  NA   1 0.1340870  2.7012367  0.91542919  13.88484
## 4     4   NA  NA   1 0.6056674  0.4534444 -0.47466093  11.19012
## 5     5   NA  NA   1 0.9705704  0.5548727 -0.59809546  11.89792
## 6     6   NA  NA   1 0.6317712  2.7220056  0.11894736  14.10450

res <- remlf90(phenotype ~ 1 + X.x,
               genetic = list(model = 'add_animal',
                              pedigree = dat[, 1:3],
                              id      = 'self'),
               data = dat)

str(ranef(res)$genetic)

## Named num [1:1020] -2.3283 0.8845 2.4141 0.0111 -0.1477 ...
## - attr(*, "names")= chr [1:1020] "1" "2" "3" "4" ...
## - attr(*, "se")= Named num [1:1020] 0.461 0.451 0.454 0.457 0.452 ...
## ..- attr(*, "names")= chr [1:1020] "3" "4" "5" "6" ...

```

Important issue Having random effects with missing values in **combination** with genetic models, can yield spurious predictions of Breeding Values. This is due to the higher variability of the residual term, for the individuals with missing values in random effects.

Missing values in coordinates of spatial effects

Are allowed. Just like in any other random effect. For those cases, the spatial component will not participate in the prediction.

```

dat <- breedR.sample.phenotype(
  fixed = c(mu = 10, x = 2),
  spatial = list(model      = 'AR',
                 grid.size = c(10, 5),
                 rho        = c(.2, .8),
                 sigma2_s   = 1)
)
dat$Var1[1] <- NA
head(dat)

##    X.mu      X.x Var1 Var2    spatial      resid phenotype
## 1     1 0.5744860   NA    4  0.5273233 -1.1815221  10.494773
## 2     1 0.8281927    6    3  0.3750826 -1.4210298  10.610438
## 3     1 0.1513396    1    1 -0.7972194  0.1066011   9.612061
## 4     1 0.8768253    3    1  0.3626787  1.8304560  13.946785
## 5     1 0.9540943    4    4 -1.2498075 -0.5273166  10.131064
## 6     1 0.1794711   10    2 -0.3797537 -0.6703586   9.308830

```

```
res <- remlf90(phenotype ~ 1 + X.x,  
              spatial = list(model = 'AR',  
                             coord = dat[, c('Var1', 'Var2')],  
                             rho    = c(0.2, 0.8)),  
              data = dat)  
  
sum(model.matrix(res)$spatial[1,])  
  
## [1] 0
```